



BOARD QUESTION PAPER: JULY 2022

MATHEMATICS AND STATISTICS

Time: 3 Hrs.

Max. Marks: 80

General instructions:

The question paper is divided into **FOUR** sections.

- (1) **Section A:** Q.1 contains **Eight** multiple choice type of questions, each carrying **Two** marks.
Q.2 contains **Four** very short answer type questions, each carrying **One** mark.
- (2) **Section B:** Q.3 to Q. 14 contains **Twelve** short answer type questions, each carrying **Two** marks. (Attempt any **Eight**)
- (3) **Section C:** Q.15 to Q. 26 contain **Twelve** short answer type questions, each carrying **Three** marks. (Attempt any **Eight**)
- (4) **Section D:** Q.27 to Q. 34 contain **Eight** long answer type questions, each carrying **Four** marks. (Attempt any **Five**)
- (5) Use of log table is allowed. Use of calculator is not allowed.
- (6) Figures to the right indicate full marks.
- (7) Use of graph paper is not necessary. Only rough sketch of graph is expected.
- (8) For each multiple choice type of question, it is mandatory to write the correct answer along with its alphabet, e.g. (a)..... / (b)..... / (c)..... / (d)....., etc.
No marks shall be given, if **ONLY** the correct answer or the alphabet of correct answer is written. Only the first attempt will be considered for evaluation.
- (9) Start answer to each section on a new page.

SECTION – A

Q.1. Select and write the correct answer for the following multiple choice type of questions: [16]

- (i) The negation of $(p \vee q) \wedge r$ is _____.
(a) $(\sim p \wedge q) \wedge r$ (b) $(\sim p \wedge q) \vee r$
(c) $(\sim p \wedge q) \vee \sim r$ (d) $(\sim p \vee q) \wedge \sim r$ (2)
- (ii) $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) =$ _____.
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$ (2)
- (iii) If $|\vec{a}|=3$ and $|\vec{b}|=4$, then value of λ for which $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{a} - \lambda\vec{b}$ is _____.
(a) $\pm \frac{9}{16}$ (b) $\pm \frac{3}{4}$ (c) $\pm \frac{16}{9}$ (d) $\pm \frac{4}{3}$ (2)
- (iv) The equation of plane passing through (2, -1, 3) and making equal intercepts on the co-ordinate axes is _____.
(a) $x + y + z = 1$ (b) $x + y + z = 2$
(c) $x + y + z = 3$ (d) $x + y + z = 4$ (2)
- (v) The equation of tangent to the curve $y = 1 - e^{\frac{x}{2}}$ at the point of intersection with Y-axis is _____.
(a) $x + 2y = 0$ (b) $2x + y = 0$
(c) $x - y = 2$ (d) $x + y = 2$ (2)
- (vi) The area of the region bounded by the curve $y = \sin x$, X-axis and lines $x = 0$, $x = \frac{\pi}{2}$ is _____ sq. units.
(a) 2 (b) 3 (c) 4 (d) 1 (2)





(vii) The differential equation of $y = c^2 + \frac{c}{x}$ is _____.

(a) $x^4 \left(\frac{dy}{dx} \right)^2 - x \frac{dy}{dx} = y$

(b) $\frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$

(c) $x^3 \left(\frac{dy}{dx} \right)^2 + x \frac{dy}{dx} = y$

(d) $\frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$ (2)

(viii) If the mean and variance of a Binomial distribution are 18 and 12 respectively then value of n is _____.

(a) 36

(b) 54

(c) 18

(d) 27

(2)

Q.2. Answer the following questions:

[4]

(i) If the statement p , q are true statements and r , s are false then determine the truth value of $(p \rightarrow q) \vee (r \rightarrow s)$.

(1)

(ii) Find the direction cosines of the vector $\hat{i} + 2\hat{j} - 2\hat{k}$.

(1)

(iii) Evaluate: $\int \frac{1}{x \log x} dx$.

(1)

(iv) Write the degree of the differential equation

$$\frac{dy}{dx} + \frac{3xy}{\frac{dy}{dx}} = \cos x$$

(1)

SECTION – B

Attempt any EIGHT of the following questions:

[16]

Q.3. Without using truth table prove that:

$$\sim(p \vee q) \vee (\sim p \wedge q) \equiv \sim p$$

(2)

Q.4. Find the inverse of the matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$

(2)

Q.5. Find the principal solutions of $\sin \theta = \frac{1}{2}$.

(2)

Q.6. Find k if one of the lines given by $6x^2 + kxy + y^2 = 0$ is $2x + y = 0$

(2)

Q.7. Show that the points $A(4, 5, 2)$, $B(3, 2, 4)$ and $C(5, 8, 0)$ are collinear.

(2)

Q.8. Find the cartesian equation of the line passing through the point $A(2, 1, -3)$ and perpendicular to the vectors $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$.

(2)

Q.9. If $y = \tan^{-1} \left(\frac{8x}{1-15x^2} \right)$ then find $\frac{dy}{dx}$.

(2)

Q.10. Evaluate: $\int \frac{1}{25-9x^2} dx$

(2)

Q.11. Evaluate: $\int_{-3}^3 \frac{x^3}{9-x^2} dx$

(2)

Q.12. Find the area of the region bounded by the curve $y = x^2$ and the line $y = 9$.

(2)

Q.13. A particle is moving along the X-axis. Its acceleration at time t is proportional to its velocity at that time. Find the differential equation of the motion of the particle.

(2)

Q.14. In a meeting 70% of the members favour and 30% oppose a certain proposal. A member is selected at random and we take $X = 0$ if he opposes, and $X = 1$ if he is in favour. Find $E(X)$ and $Var(X)$.

(2)





SECTION – C

Attempt any EIGHT of the following questions:

[24]

Q.15. Examine whether the statement pattern $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$ is a tautology, contradiction or contingency. (3)

Q.16. In $\triangle ABC$, with usual notations prove that $a^2 = b^2 + c^2 - 2bc \cos A$. (3)

Q.17. In $\triangle ABC$, $A = 45^\circ$, $B = 60^\circ$, then find the ratio of its sides. (3)

Q.18. Find the volume of the tetrahedron whose vertices are $A(-1, 2, 3)$, $B(3, -2, 1)$, $C(2, 1, 3)$ and $D(-1, -2, 4)$. (3)

Q.19. Find the angle between two lines:

$$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 2\hat{j} + \hat{k}) \quad \text{and} \quad \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda'(\hat{i} + 2\hat{j} + 2\hat{k}) \quad (3)$$

Q.20. Find the vector equation of the plane passing through the points $A(-2, 7, 5)$ and parallel to the vectors $4\hat{i} - \hat{j} + 3\hat{k}$ and $\hat{i} + \hat{j} + \hat{k}$. (3)

Q.21. Find the derivative of $\cos^{-1}x$ w.r.t. $\sqrt{1-x^2}$. (3)

Q.22. If $f(x) = 3x + \frac{1}{3x}$ find the values of x for which function $f(x)$ is decreasing. (3)

Q.23. Evaluate: $\int e^{\sin^{-1}x} \left(\frac{x + \sqrt{1-x^2}}{\sqrt{1-x^2}} \right) dx$ (3)

Q.24. Solve the differential equation $(x + 1) \frac{dy}{dx} - 1 = 2e^{-y}$. Also find particular solution when $y = 0, x = 1$. (3)

Q.25. Find expected value and variance of X , where X is number obtained on the uppermost face when a fair die is thrown. (3)

Q.26. It is known that 10% of certain articles manufactured are defective. What is the probability that in a random sample of 12 such articles 9 articles are defective? (3)

SECTION – D

Attempt any FIVE of the following questions:

[20]

Q.27. Solve the following system of equations by method of inversion.

$$x + y + z = -1, y + z = 2, x + y - z = 3. \quad (4)$$

Q.28. $\triangle OAB$ is formed by the lines $x^2 - 4xy + y^2 = 0$ and the line $2x + 3y - 1 = 0$. Find the equation of the median of the triangle drawn from origin O . (4)

Q.29. If $\vec{a}, \vec{b}, \vec{c}$ are three non co-planar vectors, then prove that any vector $\vec{r}$ in the space can be uniquely expressed as a linear combination of $\vec{a}, \vec{b}, \vec{c}$. (4)

Q.30. Solve the following L.P.P. graphically

$$\text{Maximize } z = 4x + 3y$$

$$\text{Subject to } 3x + y \leq 15,$$

$$3x + 4y \leq 24,$$

$$x \geq 0, y \geq 0$$

(4)

Q.31. If $y = f(x)$ is a differentiable function of x on interval I and y is one-one, onto and $\frac{dy}{dx} \neq 0$ on I . Also

$$\text{if } f^{-1}(y) \text{ is differentiable function on } f(I) \text{ then prove that: } \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} \text{ where } \frac{dy}{dx} \neq 0$$

Hence find the derivative of the inverse of function $y = 2x^3 - 6x$.

(4)





Q.32. Prove that: $\int \sqrt{a^2 - x^2} \, dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + c.$ (4)

Q.33. The profit function $p(x)$ of a firm selling x items per day is given by $p(x) = (150 - x)x - 1625$. Find the number of items the firm should manufacture per day to get maximum profit. Also find the maximum profit. (4)

Q.34. Evaluate: $\int_0^a \frac{1}{x + \sqrt{a^2 - x^2}} \, dx$ (4)

Renaissance Classes

